

Project TwoPager | EuroTeQaThon VI

COVER SHEET - PROJECT DETAILS

Team name: OuroNova
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What is the problem you're tackling?

A homeowner planning a kitchen renovation has no way to see what reclaimed materials would look like in their own space, no simple route to the grants they are entitled to, and no idea that the contractor arriving to strip the old kitchen will landfill oak cabinets and copper pipes worth hundreds of euros rather than sell them: three disconnected failures that together keep France's circular renovation economy locked below 1% of its potential (ADEME, 2021; ANAH, 2025).

How do you solve it?

Contractors use SightCheck to photograph demolition materials and receive an instant AI condition report, hazard flag, and resale listing in 60 seconds; homeowners upload a room photo and see it rendered with those exact reclaimed materials, alongside a personalised subsidy summary for MaPrimeRénov', CEE, and ANAH and a biophilic plant recommendation as a feature with direct in-app purchase from vetted vendors.

The result is a closed circular loop: waste becomes verified inventory, homeowners choose it because they can see it and afford it through subsidies, and every confirmed choice plants a GPS-trackable tree via Reforest'Action. Revenue is pay-per-use: €8 per contractor assessment, €4 per homeowner visualisation, plus contextual referral fees from green product partners.

THE TEAM

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I. Complexity of the problem

France produces 227 million tonnes of construction and demolition waste every year, around 70% of all national waste (ADEME, 2021). The headline recovery rate of 67% is misleading: almost all of it is crushed into low-grade backfill aggregate. Selective reuse of intact building elements such as oak cabinets, copper pipes, and stone worktops sits below 1% of demolition value, while the building sector drives 25% of national CO₂ emissions and 44% of energy consumption.

Three structural failures explain why this persists. First, residential contractors have no affordable material assessment tool: the regulated PEMD diagnostic costs €1,500 to €5,000 and applies only to buildings over 1,000 m² (AGEC, 2022), so salvageable materials default to landfill. Second, homeowners cannot visualise reclaimed materials in their own space and buy new by default, even when lower-carbon alternatives exist at comparable cost. Third, France's renovation subsidy system remains deeply inaccessible: over €1.7 billion in MaPrimeRénov' disbursements went unclaimed in 2024 because high administrative complexity and a 20% drop in certified contractors left eligible homeowners unable to navigate the process (ANAH, 2025). The three failures compound each other: without assessed inventory, homeowners cannot build a compliant renovation scope; without a compliant scope, they cannot claim subsidies; without subsidies, they cannot afford sustainable choices. Existing platforms address each failure in isolation. Backacia and Cycle Up offer reclaimed-materials marketplaces but rely on slow manual expert assessment. Houzz provides room visualisation but only for new virgin products. Government subsidy portals exist but are completely disconnected from any renovation planning tool. OuroNova is the only platform that closes the full loop.

II. Impact on the SDGs

SDG 12 (Responsible Consumption and Production): SightCheck targets 200 monthly contractor assessments in Year 1, diverting salvageable materials from landfill into verified resale inventory and unlocking each homeowner's subsidy path: OuroNova generates a personalised MaPrimeRénov' and CEE eligibility summary from the renovation scope, turning a bureaucratic barrier into a one-page output.

SDG 11 (Sustainable Cities): The room visualiser uses a Stable Diffusion img2img pipeline with ControlNet depth-preservation to show homeowners their actual space rendered with reclaimed materials, making the circular choice emotionally real rather than abstract. A 25 km default transaction radius keeps material flows hyperlocal, reducing transport emissions and strengthening neighbourhood circular loops.

SDG 15 (Life on Land): The Biophilic Intelligence layer recommends specific plants based on room orientation, ventilation, and VOC load from new materials: a peace lily for poorly ventilated rooms, a snake plant for bedrooms, a herb wall for south-facing kitchens. These appear inside the room render itself, so the homeowner sees a complete living vision, not a list of suggestions, and can purchase any recommended plant directly from vetted vendors through our platform. Every confirmed circular material choice plants a GPS-trackable tree via Reforest'Action.

Trade-offs and mitigations: Collection transport can partially offset reuse CO₂ savings; SightCheck displays the carbon cost of pickup and enforces a 25 km default radius. Pre-1990 materials carry asbestos and lead-paint risk; the AI flags these automatically and excludes high-risk categories in Phase 1. Renters, approximately 36% of French households, have limited renovation rights and sit outside our primary flows; we address this through partnerships with social landlords for shared-space assessments.

III. Feasibility & Viability

Technology: A functional prototype showing the full SightCheck and homeowner journey is live at ouro-nova.vercel.app. The flow covers photo upload, condition report (Grade A/B/C), hazard flag, resale price estimate, before and after room render with reclaimed materials and biophilic plant recommendation, and a subsidy eligibility summary, with representative outputs reflecting what the production pipeline will produce. The architecture is integration-ready: SightCheck uses a proprietary vision model for material classification and hazard detection; the room visualiser uses Stable Diffusion / FLUX img2img with ControlNet edge-detection; the Biophilic Intelligence module combines room metadata with visual analysis against a curated species database; subsidy matching uses the free Mes Aides Réno government API. The team holds an AI developer, making full AI integration a near-term engineering task.

Financial model: Contractors pay €8 per report, a clear return against disposal costs they already absorb. Homeowners pay €4 per visualisation plus a subsidy eligibility summary, a one-off cost for a one-off decision. Green product partners pay contextual referral fees. Recommended plants are purchasable directly through the platform from vetted vendors; at a 20% commission on an average €35 basket and a 15% homeowner conversion rate, this adds approximately €1k MRR by month 6, scaling with the homeowner base. An 8% marketplace transaction fee activates in Phase 2. Phase 1 targets €3k to €4k MRR from contractor reports, homeowner visualisations, and three brand partner retainers. Phase 2 marketplace fees bring projected MRR to €8k to €10k at month 12 and break-even at month 18.

Roadmap: Phase 1 (M1 to M6): SightCheck, room visualiser, and subsidy summary live in Île-de-France; 3 founding green-brand partners; 200 contractor assessments. Phase 2 (M6 to M18): circular marketplace with 50 contractors and 200 buyers. Phase 3 (Year 2 onwards): Lyon, Brussels, Berlin; carbon-credit reporting for municipalities; material-flow data licensed to urban planners. The model transfers across the EU because equivalent EPR regulation now exists in Belgium, Germany, and the Netherlands, creating the same unmet need in every target market.

Funding and next steps: Seeking €80k to €150k non-dilutive seed via ADEME, Bpifrance French Tech Seed, and EIT Climate-KIC alongside founding partner contracts. Immediate next steps: 10-contractor pilot in Île-de-France, 2 to 3 signed green brand agreements, and Mes Aides Réno subsidy API integration.

